

PAVLODAR, Kazakhstan

WMO: 360030

Lat: 52.2003N

Lon: 77.0703E

Elev: 400

StdP: 14.48

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-25.3	-37.9	0.6	-31.1	-31.2	1.0	-24.9	27.4	19.8	24.2	17.8	3.6	0	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	91.4	64.9	87.9	64.4	84.4	63.6	68.9	82.4	67.3	80.7	65.9	79.2	8.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	92.7	73.5	62.8	86.8	72.5	61.0	81.5	71.1	33.4	82.5	32.1	80.3	30.9	79.0	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.7	20.6	18.0	DB	-33.8	98.0	8.1	3.2	-39.6	100.3	-44.4	102.2	-48.9	103.9	-54.8	106.2	(4)
(5)			WB	-34.0	72.0	8.0	1.9	-39.8	73.4	-44.5	74.5	-48.9	75.6	-54.8	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	0.5	5.0	21.2	43.9	56.5	67.1	69.7	66.1	54.3	40.5	22.0	8.0	(6)
(7)		DBStd	26.84	15.15	13.37	12.91	10.04	9.05	7.29	6.12	6.95	8.46	9.13	13.96	13.76	(7)
(8)		HDD50	6473	1534	1261	895	232	42	1	0	1	50	316	839	1302	(8)
(9)		HDD65	10267	1999	1681	1359	632	288	61	25	71	336	759	1289	1767	(9)
(10)		CDD50	2119	0	0	1	50	244	514	610	500	178	22	0	0	(10)
(11)		CDD65	440	0	0	0	1	25	124	170	106	14	0	0	0	(11)
(12)		CDH74	5322	0	0	0	41	461	1492	1773	1285	266	4	0	0	(12)
(13)		CDH80	1938	0	0	0	7	136	565	667	474	89	0	0	0	(13)
(14)	Wind	WSAvg	7.8	7.3	7.7	8.7	9.0	8.6	7.4	6.9	7.0	7.1	7.7	8.3	8.2	(14)
(15)	Precipitation	PrecAvg	11.1	0.6	0.5	0.5	0.6	1.0	1.4	2.1	1.2	0.9	0.8	0.6	0.6	(15)
(16)		PrecMax	15.3	1.0	1.0	1.1	1.8	4.3	3.0	4.9	2.9	1.9	2.7	1.1	1.1	(16)
(17)		PrecMin	6.0	0.1	0.1	0.1	0.0	0.3	0.2	0.4	0.4	0.1	0.1	0.3	0.3	(17)
(18)		PrecStd	2.4	0.3	0.2	0.3	0.5	0.9	0.8	1.2	0.7	0.5	0.6	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.3	35.4	57.0	79.4	90.4	94.6	95.3	95.3	89.7	72.5	52.4	36.0	(19)
(20)			MCWB	32.8	32.6	44.4	56.7	62.5	64.1	67.2	66.3	61.6	53.2	44.7	33.6	(20)
(21)		2%	DB	31.7	32.2	48.1	72.3	83.4	90.0	90.9	89.7	81.1	66.1	46.4	33.8	(21)
(22)			MCWB	29.9	30.1	40.7	53.6	59.1	63.8	66.1	65.2	58.7	50.8	41.2	31.7	(22)
(23)		5%	DB	27.0	28.8	41.2	67.0	79.1	86.4	86.9	85.4	75.4	60.1	41.9	30.6	(23)
(24)			MCWB	25.0	27.0	36.9	50.9	57.5	63.6	65.6	64.2	56.5	48.4	37.9	28.8	(24)
(25)		10%	DB	22.8	24.9	36.9	62.1	74.6	82.5	83.4	81.2	70.0	54.4	38.3	27.6	(25)
(26)			MCWB	21.2	23.3	33.7	48.6	56.4	62.5	64.9	62.5	54.7	45.7	35.2	26.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.9	33.1	47.1	57.8	64.3	69.3	71.7	69.9	63.8	55.3	46.3	34.1	(27)
(28)			MCDWB	34.6	34.6	54.9	75.1	83.2	84.2	85.9	85.5	82.7	66.5	50.7	35.4	(28)
(29)		2%	WB	29.8	30.3	41.4	54.9	61.8	67.2	69.7	67.7	60.7	52.1	41.4	31.8	(29)
(30)			MCDWB	31.6	32.1	47.6	69.7	78.4	82.0	82.2	82.9	76.6	63.4	45.7	33.4	(30)
(31)		5%	WB	25.4	27.0	37.2	52.1	59.6	65.7	68.1	65.8	58.2	49.4	38.1	28.9	(31)
(32)			MCDWB	27.0	28.8	41.1	64.2	74.4	80.2	80.5	80.5	72.2	58.6	41.6	30.6	(32)
(33)		10%	WB	21.1	23.4	33.8	49.4	57.5	64.3	66.8	64.1	55.8	46.5	35.3	25.9	(33)
(34)			MCDWB	22.7	25.0	36.5	60.7	71.6	77.9	78.8	78.6	67.8	54.1	38.1	27.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.7	18.5	18.0	23.0	24.3	22.9	21.2	23.2	22.9	19.1	14.7	16.7	(35)
(36)			MCDBR	17.0	16.9	20.5	31.3	31.2	29.0	26.9	29.7	32.0	28.5	16.9	17.0	(36)
(37)			MCWBR	16.3	15.7	14.4	16.3	13.4	11.5	9.9	11.5	14.3	15.7	12.8	16.2	(37)
(38)		5% WB	MCDWR	17.2	17.1	19.8	28.7	27.1	24.0	22.5	25.7	28.2	25.1	15.7	17.0	(38)
(39)			MCWBR	16.5	16.1	14.7	15.9	13.1	11.0	9.5	11.0	13.7	15.0	12.4	16.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.304	0.340	0.396	0.419	0.422	0.427	0.451	0.430	0.382	0.355	0.342	0.314	(40)
(41)		taud		2.019	1.942	1.936	2.122	2.168	2.227	2.166	2.199	2.279	2.255	2.123	2.006	(41)
(42)		Ebn at Noon		204	231	243	257	263	263	253	252	251	230	189	173	(42)
(43)		Edh at Noon		26	38	47	44	44	42	44	41	33	28	24	22	(43)
(44)	All-Sky Solar Radiation	RadAvg		401	764	1196	1501	1913	2027	1830	1621	1180	647	326	261	(44)
(45)		RadStd		38	44	91	113	136	89	116	98	99	86	36	38	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page